Received: 24 June 2025 | Accepted: 11 November 2025

DOI: 10.1111/mice.70169

RESEARCH ARTICLE

COMPUTER-AIDED CIVIL AND INFRASTRUCTURE ENGINEERING WILEY

A spatiotemporal prediction method for the evolution of pavement distress in road networks

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Funding information
National Natural Science Foundation of China, Grant/Award Number: 52372305; Scientific Research Innovation Program of Shanghai Municipal Education Commission, Grant/Award Number: 2021-01-07-00-07-E00092; UK Research and Innovation, Grant/Award Number: EP/Y024257/1; Engineering and Physical Sciences Research Council, Grant/Award Number: EP/Y024257/1

Abstract

Existing pavement performance prediction models often struggle to capture complex spatiotemporal dependencies in road networks due to reliance on empirical rules and scenario-based calibration. This study proposes pavement graph network (PaveGNet), a spatiotemporal graph network framework designed to model fine-grained pavement distress evolution. It constructs a multi-node graph encoding topological correlations and time-based state transitions, while integrating exogenous factors such as temperature, precipitation, traffic, and maintenance works. Experiments demonstrate that PaveGNet performs effectively in predicting fine-grained indicators of distress evolution. The prediction error for distress evolution rate was significantly reduced, from 9.005% with the baseline spatial–temporal graph convolutional network model to 2.670%. Ablation experiments were conducted to verify the contribution of temporal interdependence, spatial correlation, and external variables in the proposed PaveGNet framework. The results demonstrate that all three components play essential roles in prediction, with external variables showing the most significant impact. To further assess the modular robustness, parts of the spatial and temporal learning modules were independently replaced. The results indicate that the prediction of distress evolution rate relies heavily on the originally designed learning components. Overall, this framework provides a more realistic and scalable solution for the spatiotemporal prediction of pavement distress evolution in road networks.

1 | INTRODUCTION

Understanding and forecasting pavement deterioration is essential for effective infrastructure management (Askarzadeh et al., 2025) and investment planning (Harvey et al., 2020; Javadinasab Hormozabad et al., 2021). In the United Kingdom, recent reports indicate that the one-time "catch-up" cost to bring local roads to a satisfactory condition has risen to £16.8 billion in 2024/2025, with estimated backlog clearance times of 12 years in England, 10 years in London, and 13 years in Wales, even under sufficient funding. In 2024/2025 alone, approximately 1.9 million potholes were filled across England and Wales at a cost of £137.4 million, equivalent to repairing one pothole every 17 s, illustrating the heavy operational pressure and reactive nature of current maintenance practices (Foundation, 2025).

Pavement distress evolution exhibits both local spatial correlations and temporal dependencies. Short-interval, high-resolution observations at individual distress points can provide actionable insights for targeted repair planning. While traditional approaches often focus on small-scale road sections, there is a growing need to extend predictive models to large-scale road networks to better capture the spatial and temporal dynamics of pavement distress. Although current methods for pavement distress detection (Hu et al., 2025) have achieved remarkable progress, such as automation in diagnosis and treatment (X. Yang et al., 2024) and real-time detection based on edge-cloud computing (Y. C. Liu et al., 2025), early efforts in predicting the evolution of such distresses primarily rely on deterministic models grounded in empirical or mechanistic principles. Widely adopted models, such as the serviceability equations developed in the United States (Abaza et al., 2001) and the World Bank's HDM-4 model (Sanchit et al., 2021), establish functional relationships between cumulative axle loads and pavement condition indicators, including roughness and rutting. Despite their widespread adoption, these methods typically rely on empirical assumptions and require extensive calibration for individual segments. The struggle to incorporate diverse external factors, such as weather variability (R. K. Yang et al., 2025) and fluctuating traffic volumes, makes them impractical for network-level deployment (J. Zhang et al., 2016).

Temporal modeling techniques, including autoregressive moving average (Chu & Durango-Cohen, 2007), autoregressive integrated moving average (Deng & Shi, 2024), and Kalman filtering (Haddad et al., 2023), have also been employed to predict long-term pavement trends based on historical observations. But their temporal-only focus often leads to biased outcomes when spatial interactions and external disruptions are not sufficiently accounted for.

Recognizing the inherent uncertainties in pavement deterioration, especially at the network scale, researchers have since turned to probabilistic and fuzzy logic-based frameworks (J. Q. Zhang et al., 2024). These methods, such as gray system theory (Shen et al., 2004), probabilistic approaches, and fuzzy logic models (Kaur & Tekkedil, 2000), have collectively formed a foundational basis for pavement performance prediction at the network level (Z. Li et al., 2024). By extracting correlations from limited datasets, such models aim to construct predictive frameworks under conditions of data scarcity (K. C. P. Wang et al., 2007). For example, Markov decision processes have been applied to simulate the temporal dynamics of urban pavement systems (Al Aryani et al., 2018), sometimes in combination with gray theory (J. M. Wu & Dai, 2010) or empirical adjustments (Abaza, 2017) to improve prediction accuracy. However, these methods are still constrained by their reliance on minimal prior information (Abaza, 2016) and often rest on overly simplistic assumptions (Bianchini & Bandini, 2010), such as state independence or weak spatial–temporal interdependencies, that limit their ability to capture the interconnected nature of pavement degradation processes across large and diverse road networks (H. Zhang et al., 2024). These methods with underlying assumptions often deviate from real-world complexity due to a lack of real data support. Therefore, these approaches are generally more suitable for performance forecasting on a small number of road segments with limited data availability.

With the rise of data availability (Z. Y. Yang et al., 2025) and computational power, data-driven methodologies have shown promising potential in addressing the complexities of spatiotemporal pavement prediction (Kutz et al., 2016). Leveraging data collected from multi-sensor platforms (Askarzadeh et al., 2025), mobile monitoring systems (Shahini Shamsabadi et al., 2015), and crowdsourced inputs, these approaches enable more granular, real-time assessment of infrastructure health (Amezquita-Sanchez et al., 2018). Large-scale implementations have demonstrated success in detecting early deterioration patterns, optimizing maintenance prioritization (Kitaha & Biligiri, 2017), and integrating life cycle and environmental considerations into decision support frameworks (Xin et al., 2023). To analyze rich data, artificial neural networks, such as neural dynamic classification algorithm (Rafiei & Adeli, 2017), dynamic ensemble learning algorithm (Alam et al., 2020), and hybrid models have gained traction (J. Yang et al., 2003). These methods exploit the non-linear mapping capabilities of deep learning algorithms (Rafiei & Adeli, 2018) to identify patterns in pavement performance data. Researchers have further integrated neural networks (Xiao et al., 2023) with wavelet transforms (G. W. Yang et al., 2018), Monte Carlo simulations (Dao et al., 2020), physics-informed mechanisms (Y. X. Wu et al., 2025), and Markov chains (Z. Wang et al., 2021) to improve prediction robustness. Moreover, researchers have also incorporated uncertainty into neural network-based predictive models to enhance the reliability of their forecasts (Perez-Ramirez et al., 2019). However, despite these enhancements, the effectiveness of such models heavily depends on the availability of high-quality historical data and remains sensitive to noise, missing values, and inconsistent sensing environments (Y. Du et al., 2016). Existing mainstream datasets, such as the long-term pavement performance database, primarily rely on aggregated performance indices, such as the pavement condition index (PCI) and international roughness index, which are derived from weighted combinations of various distress types. While useful for broad assessments, these indices often obscure critical details about individual distress manifestations. As a result, access to high-quality datasets that retain raw, disaggregated distress information is crucial for developing robust and interpretable predictive models. Besides that, existing models, whether traditional, probabilistic, or data-driven, tend to simplify the complex, multidimensional interplay among deterioration factors or treat individual road segments in isolation, overlooking the interconnected nature of road networks (Ilicali et al., 2014). This limited perspective often fails to capture the spatial dependencies between segments, the temporal continuity embedded in historical degradation patterns, and the stochastic influence of external disruptions such as extreme weather events or abrupt maintenance actions. Furthermore, the computational burden and high data quality requirements of many advanced models continue to pose obstacles to their scalability and practical deployment across large, heterogeneous networks.

Graph, as an important data structure, is composed of a series of nodes and the edges that connect these nodes. To address real-world problems that can be abstracted as graphs (Ahmadlou & Adeli, 2017), Franco proposed the graph neural network (GNN) based on Banach's fixed point theorem (Zhou et al., 2020). GNN can address spatial problems with a clearly defined graph structure. However, scenarios such as Euclidean spaces (such as images) or sequences (such as textual information) can also be abstracted into graph problems for modeling and analysis using GNNs (F. Y. Liu & Al-Qadi, 2025). With the advancement of convolutional neural networks (CNNs), Bruna first proposed two types of CNNs on graphs in 2013, which are spectral domain-based and spatial domain-based CNNs, building on graph signal processing (Bruna et al., 2013). Spatial domain-based CNNs, due to their relatively intuitive physical interpretation, have gained significant attention from researchers and are better suited for analyzing the evolution of pavement distress. However, they remain limited in their ability to capture long-term dependencies and complex temporal dynamics inherent in road network deterioration.

In recent years, several state-of-the-art methods have been proposed for time series prediction, including classical sequential models such as recurrent neural networks (RNNs; Hewamalage et al., 2021), long short-term memory networks (LSTMs; T. Li & Wu, 2023), and Transformers (Han et al., 2022), as well as spatial–temporal graph-based approaches such as the spatial–temporal graph convolutional network (STGCN; Gomes et al., 2021). However, most of these models are designed for fixed-node networks and assume stable and complete spatial structures over time. In contrast, pavement distress data exhibiting dynamic and incomplete graph structures-nodes (representing individual distresses) may emerge, expand, or be repaired across time, resulting in continuous changes to the graph topology at each time step. This inherent non-stationarity poses significant challenges for conventional STGNNs, which rely on static adjacency matrices. Therefore, a dynamic graph construction mechanism is required in the spatial dimension to adaptively capture evolving spatial correlations among distresses based on the observed spatial configuration. In addition, due to the sparsity and irregular sampling of pavement monitoring data, it is essential to incorporate a temporal semantic reconstruction strategy to recover continuous temporal representations from incomplete observations. Furthermore, the evolution of pavement distress is influenced by cumulative and causal dependencies from past states, necessitating a modeling mechanism that can explicitly capture physically consistent temporal dependencies rather than merely smooth temporal correlations. Above all, these challenges highlight the limitations of existing models and motivate the development of a unified framework capable of learning from dynamic, sparse, and physically constrained spatial–temporal data.

To address these challenges, we propose a spatiotemporal evolution framework that integrates historical degradation trajectories, network-wide spatial correlations, and dynamic external influences into a unified prediction scheme. A spatiotemporal graph-based method (pavement graph network [PaveGNet]) is proposed that enables accurate prediction of the evolution of pavement distress across road networks. The main contributions of this study are as follows: (1) Dynamic spatial representation for evolving pavement distresses: Unlike static networks, PaveGNet dynamically constructs the spatial graph at each time step based on the observed spatial configuration of pavement distresses. This adaptive mechanism allows the model to capture the continuously changing correlations among distresses as they emerge, expand, or are repaired over time. (2) Temporal semantic reconstruction for sparse monitoring data: To address the temporal sparsity and irregular sampling of field observations, the model reconstructs temporally continuous feature sequences by leveraging contextual information from adjacent time intervals. This enables the preservation of meaningful temporal patterns even when data are incomplete or unevenly spaced. (3) Causally consistent temporal dependency modeling: The framework explicitly incorporates causal temporal dependencies to ensure that only historical and physically meaningful information contributes to the prediction process. This design prevents information leakage from future states and captures the accumulated influence of environmental and traffic-related factors on distress evolution.

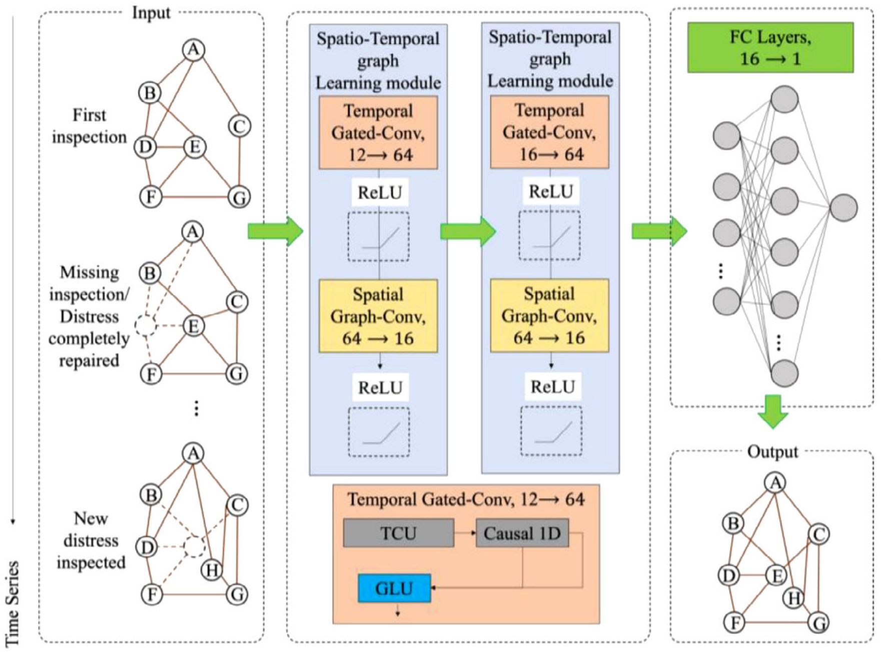

FIGURE 1 The overall framework of the pavement graph network (PaveGNet) framework. FC, fully connected; GLU, gated linear unit; TCU, temporal completion unit.

This framework can combine the consideration of the temporal deterioration of a single distress location and the impact of distress at other locations in the road network on the target distress. The predictive capability of our model is enriched by integrating a wide range of external influencing factors, including temperature, precipitation, traffic volume, and historical maintenance records. This comprehensive incorporation of environmental and operational data enables PaveGNet to accurately predict the evolution of pavement distress while maintaining physical interpretability and robustness to data incompleteness.

The rest of this article is organized as follows. Section 2 introduces the PaveGNet framework for spatiotemporal prediction of pavement distress evolution and its detailed architecture. Section 3 shows the spatiotemporal pavement distress dataset for model training, validation and testing. Section 4 verifies the proposed PaveGNet framework based on the spatiotemporal pavement distress dataset and investigates the feasibility. Section 5 discusses the ablation analysis on the spatial learning modules, temporal learning modules and external variables. Section 6 summarizes the study outcomes and generates the conclusion.

2 | METHODOLOGY

2.1 | Overall methodology

To predict the evolution of pavement distress, we propose the PaveGNet framework as illustrated in Figure 1.

The architecture of the PaveGNet framework for predictive learning consists of three core processes: (1) The data preprocessing component (DPC) is responsible for extracting and constructing spatiotemporal graph data from the raw data. (2) The spatiotemporal graph learning component (STGLC) focuses on capturing the underlying spatiotemporal correlations in complex spatial systems. (3) The task-aware forecasting component maps the spatiotemporal latent features extracted by the STGLC to the corresponding domain of subsequent prediction of indicators.

In the DPC process, the input layer consists of a series of time-step observations, with each time step containing features. The spatiotemporal sequence data for pavement distress are constructed as $X = \{x_t \in R^{N\times F} | t = 0, \ldots, T\}$, where N is the number of spatial nodes representing the number of pavement distress sampling points, and this value changes with each collection time. In time series, some inspections may be missing, or certain distresses may be repaired while new ones are identified. As a result, nodes in the graph may be missing or newly added over time. F is the feature dimension, which includes parameters such as the area of cracks (both longitudinal and transverse cracks), alligator cracks, potholes, and patched areas. Considering that pavement distress evolution is closely related to environmental factors, parameters such as temperature, temperature difference, humidity, precipitation, and traffic volume data are also included as feature inputs. At each time step, the graph capturing the correlations among distresses is adaptively updated according to the spatial configuration of the observed data.

In the STGLC process, there are two spatiotemporal graph learning modules, consisting of one temporal gated convolution layer (Temporal Gated-Conv) and one spatial graph convolution layer (Spatial Graph-Conv), separately. Temporal Gated-Conv is designed to extract dynamically changing features along the time dimension. The Temporal Gated-Conv layer includes a temporal completion unit (TCU), a causal 1D convolution, and a gated linear unit (GLU), which processes the temporal dimension features of the input data. For each distress sampling point, information from its adjacent temporal domain is utilized to reconstruct representative parameters for the centric time step. This strategy addresses data gaps caused by missing or unscheduled inspections, as well as newly emerging distresses. In addition, the causal 1D convolution with GLU is incorporated to ensure that only historical and physically meaningful temporal dependencies are propagated, thereby avoiding future information leakage and capturing the accumulated historical influence of weather and traffic conditions.

The Spatial Graph-Conv layer is used to process spatial correlations. It performs graph convolution operations on the output of the Temporal Gated-Conv process. Data from the same time step are aggregated together, and different locations within the same time step are used to establish spatial associations, considering the graph structure (adjacency matrix) information.

Finally, by connecting the fully connected layers (FC layers), the predicted value of the indicator of the final target can be obtained.

2.2 | Temporal interdependence modeling

The overall Temporal Gated-Conv structure is shown in Figure 2.

Each node in the graph represents a distress sample point (A–H). Data collection for each node is a time step. Considering the newly detected distress and missed distress in each collection, the graph of each time step will have different states. The dotted nodes represent sample points that have not been detected, and the edges connected to them are dotted lines, indicating that the sample point has no spatial association with neighboring distresses at this time point. Each node contains all the feature information.

In the TCU process, for each distress sampling point, a specific time step is selected, and its adjacent temporal intervals are identified to form a temporal region of interest. For each feature within this region, a 1×1 convolution is first applied for dimensional expansion, followed by another 1×1 convolution for dimensional reduction. This process enables the integration of temporal context from neighboring time steps, facilitating the reconstruction of missing or uncollected information. The operation is then performed across all time steps. This process ultimately yields a complete and temporally enriched graph structure, in which each distress sampling point is associated with a full sequence of temporal information.

The outputs are further used to facilitate the input of subsequent causal 1D convolution and GLU gating mechanism. 1D convolution essentially fuses features from multiple time steps into a richer temporal feature representation, so it can be considered as merging information from multiple time steps. We utilize the causal 1D convolution kernel to consider only the historical feature information for the current time step, avoiding the use of future information.

In addition, the operation principle of GLU is to selectively retain and suppress input features through a gating mechanism. GLU first splits the input features into two parts by dividing the number of input channels into two halves. Next, a sigmoid activation function is applied to one part of the features, generating a gating signal between 0 and 1. This gating signal determines the relative importance of retained information and suppressed information. Where the gating signal is close to 1, it indicates that retaining information is important, meaning that the time-dimension relationship of relevant distress is significant. Where the gating signal is close to 0, it indicates that suppressing information is important, meaning that the time-dimension relationship of relevant features is not significant. The gating signal is multiplied by the other part of the features, combining the retained and suppressed information. This operation allows the network to selectively enhance or weaken features to autonomously learn important distress evolution information. In the case of pavement performance evolution analysis, it can autonomously select environmental factors that have an impact on distress evolution. Each node uses this time series model to process its time series features independently and finally uses them as input for spatial convolution.

We applied batch normalization (BN) and Dropout exclusively along the temporal interdependence modeling. Specifically, for each node, the feature sequences across time steps are normalized using BN after the 1×1 convolution layers, stabilizing the training by mitigating covariate shift in sequential features. Subsequently, a dropout layer (dropout rate = 0.2; Srivastava et al., 2014) is applied along the same temporal dimension to regularize the model and prevent over-reliance on specific time steps.

2.3 | Spatial correlation modeling

In the scenario of modeling the evolution of pavement distress at the road network level, nodes represent each distress, and edges represent the relationship between

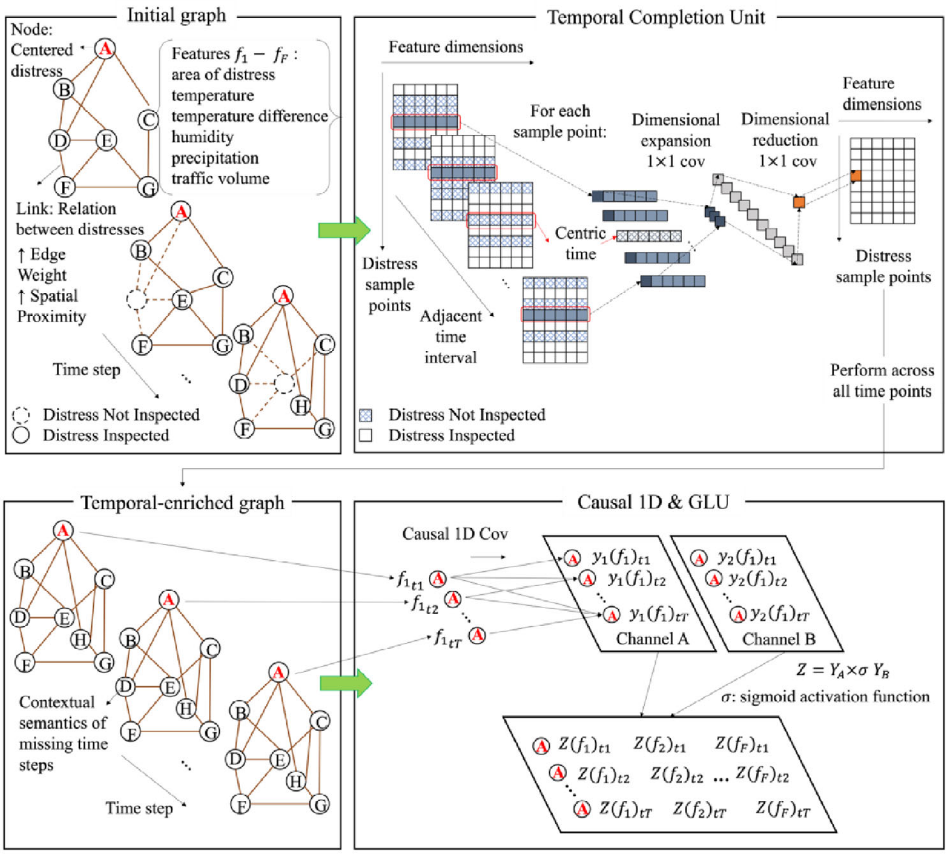

FIGURE 2 The overall Temporal Gated-Conv structure. GLU, gated linear unit.

distresses. For an undirected road network graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, which consists of N nodes (distress locations) $v_i \in \mathcal{V}$ and M edges (distress relationships) $(v_i, v_j) \in \mathcal{E}$, the adjacency matrix $A \in \mathbb{R}^{N\times N}$ can have elements represented as binary variables or weighted variables, and the degree matrix is $D_{ii} = \sum_j A_{ij}$. Each node in the graph has its own features X_v that contain internal attribute information and external influence factors. Besides, each edge also has its own features $X_{(v_i,v_j)}$, which are weights that represent the relationship between distresses. The elements in the adjacency matrix are computed using a distance-related summation function, and the adjacency matrix of the distance-based graph using the Gaussian radial basis function can be formulated as shown in Equation (1):

$$a_{ij} = \begin{cases} \dfrac{\exp\left(-\|d_{ij}\|_2\right)}{\sigma}, & if\ d_{ij} \ll \epsilon \\ 0, & otherwise \end{cases} \tag{1}$$

where d_{ij} is the distance between nodes i and j, ϵ is a predefined threshold that controls the sparsity of the adjacency matrix, and σ is a hyperparameter that controls the distribution. Therefore, if the distance is smaller, the weight is larger, indicating that the nodes are more strongly correlated, and vice versa. The spatiotemporal graph can be represented as $G_t = (V, E_t, A_t)$, where V is the set of nodes, E_t, and A_t represent the set of edges and the adjacency matrix at time t, respectively.

The core of spatial correlation modeling is to learn the graph-aware hidden state h_v, which contains both the information of the distress node itself and its neighbouring distress nodes. The hidden states of all nodes are iteratively updated. The hidden state of the node v at time $t+1$ is updated as shown in Equation (2):

$$h_v^{t+1} = g\left(X_v, X_c O[v], h_n^t e[v], X_n e[v]\right) \tag{2}$$

where g is the state update function, a globally shared feature that allows the model to capture and transmit temporal dependencies within the sequence. $X_c O[v]$ represents the edge features connected to the distress node v, $h_n^t e[v]$ represents the hidden state of the neighboring distress nodes of v at time t, and $X_n e[v]$ represents the features of the neighboring distress nodes. The state update function for the hidden state can be viewed as a key internal mechanism of the model, ensuring the effective transmission of distress evolution information and dynamic processing of the sequence. To simplify, this function performs local state transitions at each step, driving the gradual evolution of the entire sequence.

The model's convergence is defined by checking whether the norm difference between two consecutive time steps is smaller than a certain threshold, as shown in Equation (3):

$$\left\|\boldsymbol{H}^{t+1}\right\|_2 - \left\|\boldsymbol{H}^{t}\right\|_2 < \varepsilon \tag{3}$$

The state update function g is implemented as a contraction mapping through a feedforward neural network. Specifically, when processing graph-structured data, the features of each distress node and the features of all distress relationships connected to it, as well as the features and hidden states of its neighboring distress nodes, are concatenated. This concatenation operation effectively integrates the local contextual information of the distress node, providing a rich feature representation for subsequent processing, as shown in Equation (4):

$$\begin{aligned} h_v^{t+1} &= g\left(X_v, X_c O\left[v\right], h_n^t e\left[v\right], X_n e\left[v\right]\right) \\ &= \sum\nolimits_{v_j \in e[v]} FNN\left(\left[X_{v_i}; X_{(v_i, v_j)}; h_{v_j}^t; X_{v_j}\right]\right) \end{aligned} \tag{4}$$

To ensure the contractiveness of the g mapping, a penalty term targeting the Jacobian matrix is applied to restrict the size of the partial derivative matrix. Specifically, a mapping is contractive if and only if the magnitude of its gradient or derivative is less than 1. This equivalence condition can be derived from the formal definition of a contraction mapping based on the norm of the space, which ensures that the mapping's gradient magnitude remains smaller than 1. To simplify the analysis, consider the one-dimensional case, where the difference between coordinates can be treated as the distance of a vector in space, leading to the derivation of Equation (5):

$$\left|\left|g'\left(x\right)\right|\right| = \frac{\partial g\left(x\right)}{\partial x} \leq c, 0 \leq c < 1 \tag{5}$$

Equation (5) describes the mathematical properties of contraction mappings and their relationship with the magnitude of the derivatives. In this way, it ensures that distress state transitions remain stable during optimization and prevents divergence caused by irregular graph structures or noise from field measurements. The original constrained optimization problem is transformed into a more manageable unconstrained optimization problem, and based on this, a stable and effective model is trained. The training objective can be represented by Equation (6):

$$J = Loss + \lambda \cdot \max\left(\frac{\partial FNN}{\partial h} - c, 0\right), c \in (0, 1) \tag{6}$$

where $Loss$ is the model's loss function, and $\lambda = 0.01$ (Finlay et al., 2020) and $c = 0.9$ (Gouk et al., 2021) are hyperparameters, and the term multiplied by it is the penalty term for the Jacobian matrix. During the training process, the parameters are defined through supervised learning to construct the loss function. Assuming there are p supervised nodes, the model's loss function is defined as in Equation (7):

$$Loss = \sum_{i=1}^{p} \left(t_i - o_i\right) \tag{7}$$

where t_i is the target value, and o_i is the output value. According to the Almeida–Pineda algorithm (Y. Li et al., 2015), during forward propagation, the model calls g a total of T_n times, until $h_v^{T_n}$ it converges. Here, T_n represents the number of iterations required for the hidden states to satisfy the convergence criterion defined in Equation (3), when the norm difference between two consecutive time steps becomes smaller than a predefined threshold ε. This iteration process ensures that the model's hidden representations reach a stable equilibrium before computing the final outputs. For those supervised distress nodes, their hidden states are processed through a specific output layer with the output layer function k to obtain the model's predicted output. Then, based on the difference between these outputs and the true labels, the model's loss can be calculated. During backpropagation, the gradients of g and k with respect to the final hidden state $h_v^{T_n}$ can be directly computed. However, because the model recursively calls g, to compute the gradients of g and k with respect to the initial hidden state h_v^0, the gradients need to be recursively computed T_n times. The final gradient obtained is the gradient of f and g with respect to the initial hidden state h_v^0, which is then used to update the model's parameters.

The propagation rule for constructing a multi-layer graph network model is given by Equation (8):

$$\boldsymbol{H}^{l+1} = \sigma\left(\tilde{\boldsymbol{D}}^{-\frac{1}{2}} \tilde{\boldsymbol{A}} \tilde{\boldsymbol{D}}^{-\frac{1}{2}} \boldsymbol{H}^{(l)} \boldsymbol{W}^{(l)}\right) \tag{8}$$

where $\tilde{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{I}_{\mathrm{N}}$ is the adjacency matrix of the undirected graph $\mathcal{G}$ with added self-connections, where $\boldsymbol{I}_{\mathrm{N}}$ is the identity matrix. $\tilde{\boldsymbol{D}}_{ii} = \sum_j \tilde{\boldsymbol{A}}_{ij}$ and $\boldsymbol{W}^{(l)}$ is the weight matrix for the specific layer. $\sigma(\cdot)$ represents the activation function, and in this study, $ReLU(\cdot)$ is chosen, where $ReLU(\cdot) = \max(0, \cdot)$. $H^{(l)} \in \mathbb{R}^{N \times D}$ is the activation matrix of the lth layer, and $\boldsymbol{H}^0 = \boldsymbol{X}$.

As shown in Equation (9), v_i refers to the ith distress node in the graph, and c_{ij} represents the normalization coefficient. The key feature of this formula is that the feature of each distress node in the graph is updated based on the collective influence of other distress nodes' features

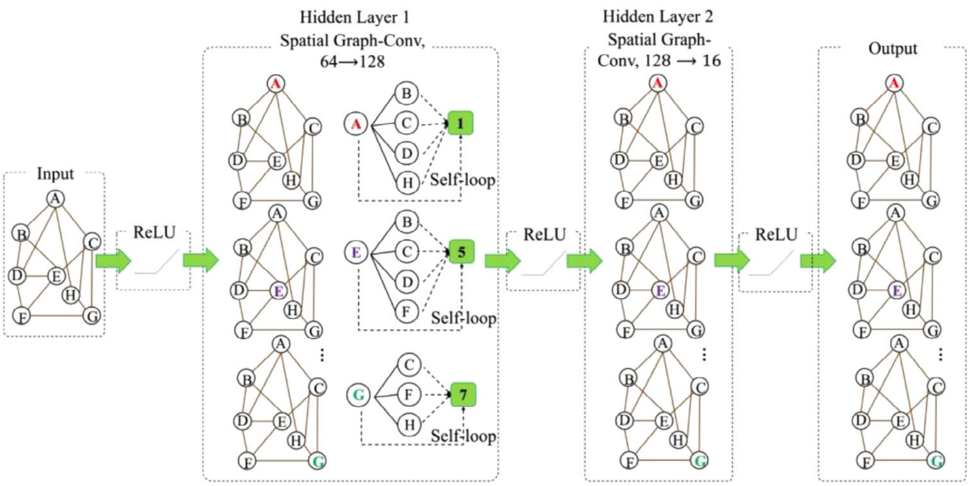

FIGURE 3 The overall spatial graph-conv structure.

processed by a hash function. Therefore, the distress node features are propagated through the adjacency matrix A, where the normalized degree matrix $\tilde{D}$ is used to normalize the node features by taking the inverse square root. The features are then updated via the weight parameters and activation function, which represent the state update of distress in the spatial domain. At the same time, $h_{v_i}^{l+1}$ represents the weighted sum of the features of the neighbouring distress nodes of node v_i in the $l+1$th layer, after being transformed by the weight parameters and activation function. This reflects the correlation between the change of a single distress point and the distresses in its neighbouring area.

$$h_{v_i}^{l+1} = \sigma\left(\sum_j \frac{1}{c_{ij}} h_{v_j}^{l} \boldsymbol{W}^{(l)}\right) \tag{9}$$

When the graph structure changes, the Laplacian matrix of the graph also needs to be recalculated. Therefore, this simplifies the graph convolution operation further by directly performing graph convolution in the spatial domain as shown in Equations (10)—(12):

$$g_w \times x = w\left(\boldsymbol{I}_{\mathrm{N}} + \boldsymbol{D}^{\frac{1}{2}}\boldsymbol{A}\boldsymbol{D}^{\frac{1}{2}}\right)x \tag{10}$$

$$h_{\mathcal{N}(u)}^{k} \leftarrow Aggregate_k\left(\left\{h_{u'}^{k-1}, \forall u' \in \mathcal{N}_k(u)\right\}\right) \tag{11}$$

$$h_u^k \leftarrow \sigma\left(W^k \cdot Concat\left(h_u^{k-1}, h_{\mathcal{N}(u)}^{k}\right)\right) \tag{12}$$

Among them, $\boldsymbol{A}$ represents the adjacency matrix, $\boldsymbol{D}$ is the degree matrix, and w is the learnable parameter. $\mathcal{N}(u)$ is the set of neighbour nodes of u, and $h_{\mathcal{N}(u)}^{k}$ represents the embedded representation of node u after the aggregation operation.

The overall Spatial Graph-Conv structure in this study is shown in Figure 3. We adopt two hidden layers in the Spatial Graph-Conv structure. The first layer takes 64-dimensional features from the Temporal Gated-Conv process. The output dimension controls the number of parameters and complexity of the model. A larger output feature dimension usually allows for better capture of the complex relationships between nodes. The second layer's input features match the output features from the previous layer, which are 128-dimensional parameters. The output is 16-dimensional, indicating that after the second hidden layer, each node will have a 16-dimensional feature representation. This layer further reduces the feature dimension, simplifying the model and extracting higher-level features. ReLU functions are applied between layers to enhance nonlinear representational capacity and facilitate model convergence (Chen & Ho, 2019). In graph convolutional networks (GCNs), each layer aggregates information from a node's immediate neighbors. After two layers, the node representation becomes:

$$\boldsymbol{H}^{(2)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\boldsymbol{X}W^{(0)}\right)\boldsymbol{W}^{(1)}\right) \tag{13}$$

which allows each node to incorporate information from its own state as well as from directly and indirectly connected neighboring distress. In the context of pavement deterioration, influences from third-order or more distant neighbors are typically very weak on the centered distress, so two layers are sufficient to model how a distress may affect its surrounding areas without needing deeper layers. In addition, using additional layers may lead to over-smoothing, where node representations become overly similar, and increase model complexity without clear benefit (X. Wu et al., 2023). Therefore, two layers strike a balance between capturing essential spatial dependencies,

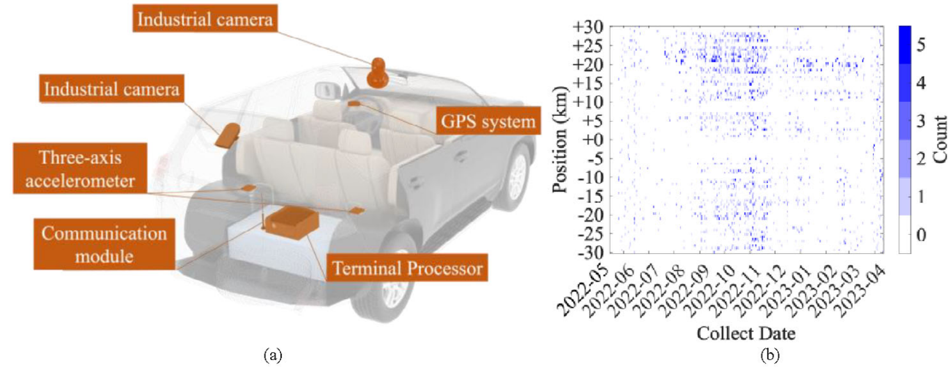

FIGURE 4 Collection equipment layout and the spatio–temporal distribution of sampling points.

maintaining feature diversity, and ensuring computational efficiency.

3 | DATA PREPARATION

To verify the proposed method, the data acquisition process lasted for 12 months from May 2022 to April 2023 and achieved the periodic collection of pavement distress data of fixed lines. The acquisition setup included high-resolution industrial cameras, a navigation unit with an integrated GPS system, and a terminal processor for real-time data logging, as shown in Figure 4a. This configuration enabled the simultaneous capture of pavement images and corresponding metadata such as timestamp, geographic coordinates, and vehicle orientation. During deployment, the industrial camera was mounted at a fixed angle on the vehicle's rear windshield to ensure consistent image capture of the pavement. Meanwhile, the GPS-integrated navigation receiver was magnetically secured to the vehicle roof, providing real-time spatial positioning and directional information. The surveyed highway section consists of asphalt road sections.

Figure 4b shows the spatiotemporal distribution of sampling points. The x-axis represents the time span, the y-axis denotes the geographic identifiers, and the color bar indicates the number of measurements collected at each location. Figure 5 illustrates samples of the raw spatiotemporal image data. Figure 5a shows a case of post-repair worsening of potholes, while Figure 5b shows a case of worsening cracks. Both primarily highlight periods of rapid local changes. Figure 5c depicts a longer-term monitoring period, which includes incomplete distress instances that were manually identified and excluded as outliers during preprocessing. After the data acquisition process, spatial coordinates, classification labels, and geometric dimensions of each distress were extracted (Y. C. Du et al., 2021) with manual re-verification. To further mitigate sharp fluctuations caused by image recognition errors, we applied a quantile-based method to identify and remove extreme outliers. Specifically, values beyond the 5th and 95th percentiles were treated as noise and corrected or removed, ensuring the dataset reflects reliable and consistent measurements. And these attributes are matched with environmental factors through timestamps. Environmental factors included meteorological data and traffic volume.

Meteorological data across the tracking period were obtained from official government platforms, containing daily average temperature, temperature difference, daily humidity, and total daily precipitation. It inherently incorporated features indicative of extreme weather conditions, including typhoons.

Traffic volume was collected from 462 mm-wave radar devices deployed on both sides of the highway. They could monitor traffic flows, including those in the upstream, downstream, and adjacent regions. The traffic volume at the section closest to a distress spot was assigned as the traffic volume corresponding to that distress. Traffic volumes were categorized into small, large, and super-large vehicles and converted to weighted equivalents using axle-load factors (0.5, 2, and 5) to account for their relative impacts on pavement deterioration. The resulting weighted sum was used as the model's traffic volume input.

Maintenance activities play a significant role in improving pavement performance. The maintenance data used in this study were collected simultaneously during pavement distress detection, ensuring completeness and traceability. Maintenance activities, such as "patched cracking," "patched alligator cracking," and "patched potholes," were identified as the primary interventions leading to post-maintenance distress improvements.

Finally, a registered distress dataset was aligned with environmental conditions and pavement distress detection results. A total of 60,526 images (1920 × 1080 pixels) with pavement distress and spatiotemporal information were obtained. After correction of distress locations and sizes, together with spatiotemporal matching (Pan et al., 2023),

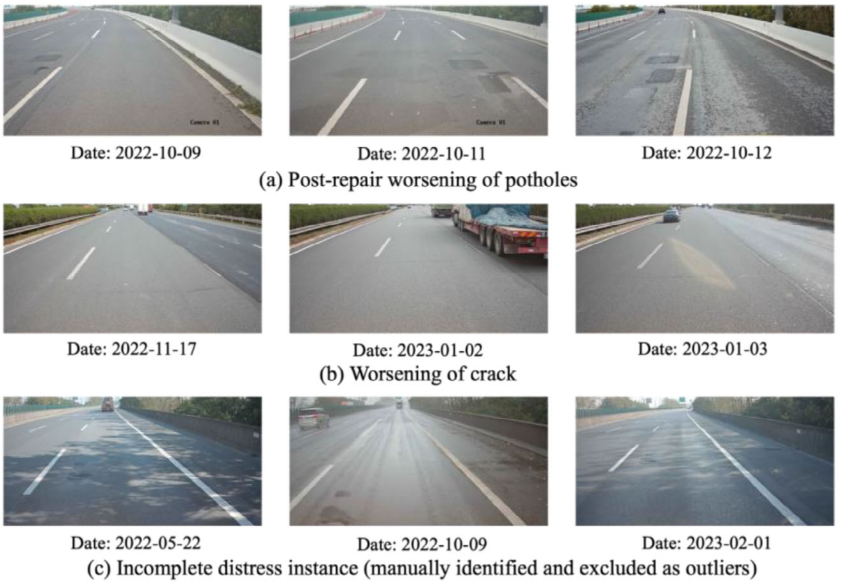

FIGURE 5 Samples of the raw spatiotemporal image data.

800 instances of spatiotemporal distress data over 247 times were tracked. Samples of the spatiotemporal dataset for pavement distress evolution are shown in Table 1.

In this framework, the features were scaled to a standard normal distribution during the training process. The objectives of normalization include:

1. Eliminating the effect of feature scale: By normalizing, all feature values are scaled to a similar range, thereby eliminating the scale effect on the model. For example, the traffic volume input feature has a very large magnitude, while the area parameters of different types of pavement distress are much smaller, and the dimensions between these two are not directly convertible.
2. Accelerating model convergence: Normalization helps the optimization process of the loss function converge more quickly. This means that the model requires fewer training iterations to achieve reasonable performance, thereby reducing training time. As the prediction of pavement distress at the road network level is a complex problem, the computation time is a key consideration.
3. Improving model stability: In some cases, the scale differences between features can lead to numerical instability, such as vanishing gradients or exploding gradients. Normalization helps reduce these numerical stability issues.

4 | RESULTS

The 800 distress locations obtained from the data preparation process served as the nodes in the PaveGNet framework, with the relationships between the distress sampling points representing the edges of the graph network. The data were divided into a training set, validation set, and test set with a ratio of 7:1.5:1.5. The prediction targets were localized PCI PCI', distress evolution rate δA, and total distress area A_{sum} indicators.

Localized PCI PCI' can effectively capture the condition of pavement distress at a localized scale. Drawing on the widely used PCI as a reference, the PCI' value for each distress instance was calculated using Equations (14) and (15):

$$PCI' = 100 - a_0 DR^{a_1} \tag{14}$$

$$DR = 100 \times \frac{\sum_{i=1}^{i_0} w_i A_i}{A} \tag{15}$$

where DR denotes the pavement distress rate (%), $a_0 = 15.00$ and $a_1 = 0.412$ are empirical coefficients specific to asphalt pavement. i is the type of distress, and i_0 is the total number of distress types. A_i represents the area (m^2) of the ith type of distress, while A refers to the convex hull area (m^2) of the distress site, calculated from a set of pixel coordinates (x_1, y_1), $(x_2, y_2), \ldots, (x_n, y_n)$ using the following Equation (16):

$$S = \frac{1}{2}\left(\sum_{i=1}^{n-1}(x_i y_{i+1} - x_{i+1} y_i) + (x_n y_1 - x_1 y_n)\right) \tag{16}$$

If multiple distress types overlap at a single observation point, they were aggregated and treated as a single combined instance for that location. w_i is the distress coefficient corresponding to the ith type of distress, determined according to distress type: 2.0 for transverse and longitudinal cracks, 1.0 for alligator cracks and potholes, 0.2 for

TABLE 1 Examples of spatiotemporal information of pavement distress evolution in the dataset.

Parameters	Physical meaning	Unit	Sample 1	Sample 2	Sample 3
ID	Unique identifier	/	640	640	640
Collect date	Date of data collection	/	2022-10-09	2022-10-11	2023-02-14
Transverse crack area	Area of transverse cracking across the lane	m^2	0	0	0
Longitudinal crack area	Area of longitudinal cracking parallel to the traffic direction	m^2	0	0	0
Alligator crack area	Area of interconnected alligator cracks	m^2	0	0	0
Pothole area	Area of potholes	m^2	0.2340265	0	0
Patched crack area	Area of linear cracks patching	m^2	0	0	0
Patched alligator crack area	Area of alligator cracks patching	m^2	0	0	0
Patched pothole area	Area of pothole patching	m^2	1.5196186	2.0090298	2.9183557
Daily temperature	Daily temperature of the distress location	°C	17.4	16	17.5
Daily temperature difference	Daily temperature difference of the distress location	°C	6.1	6.9	5.7
Humidity	Humidity of the distress location	%	73	48	60
Precipitation	Precipitation of the distress location	mm	0.8	0	0
Traffic volume	Traffic volume of the distress location	Vehicle/Day	42736	48126	46323
Total area A_{sum}	Total area for all distresses	m^2	1.7536451	2.0090298	2.9183557
Kilometer stake number	Geographic identifier of the distress	km	+60.2	+60.2	+60.2

patched cracks, and 0.1 for patched alligator cracks and patched potholes.

Distress evolution rate δA indicator is designed to capture the dynamic trend of distress development as shown in Equation (17). This metric is introduced to reflect the evolution speed of pavement distress progression over time.

$$\delta A = \frac{A_{i+1} - A_i}{t_{i+1} - t_i} \tag{17}$$

where A_i represents the actual observed distress area of the ith sample, A_{i+1} corresponds to the actual observed distress area of the $(i+1)$th sample, t_i and t_{i+1} represent the respective observation dates. If data for the subsequent time step were available, the distress evolution rate was computed accordingly. Otherwise, if no subsequent data existed, the corresponding data entry was excluded from the analysis. This process yielded the magnitude of the distress evolution trend.

The total distress area A_{sum} indicator served as a control group for comparison. In the experiment, the input features for each sampling point included the areas of transverse and longitudinal cracks, alligator cracking, potholes, patched cracks, patched alligator cracking, and patched potholes. Accordingly, the total distress area A_{sum}, directly defined as the linear combination of these measured parameters, is selected as the prediction target for the control group. This indicator provided a benchmark to assess and validate the predictive performance of the proposed model.

During the training loop, the model was trained using the data from the training set. The model parameters were updated using backpropagation. The learning rate was set to 0.001. The batch size was set to 1, as aggregating multiple graphs per batch would be inappropriate given that each time step represents a distinct spatial graph. The maximum number of iterations T_n was set to 1000. Early stopping with a patience of 250 epochs was applied, monitoring the validation loss; training halts if no

TABLE 2 Performance evaluation of the proposed pavement graph network (PaveGNet) framework.

	Training			Validation			Testing		
Predicted indicators	**Loss**	**Mean absolute error (MAE)**	**Mean absolute percentage error (MAPE)/%**	**Mean squared error (MSE)**	**MAE**	**MAPE/%**	**MSE**	**MAE**	**MAPE/%**
A_{sum}	0.08809	0.13351	2.575%	0.13111	0.15227	2.781%	0.13119	0.15157	2.910%
PCI'	1.29780	0.54404	3.750%	1.07925	0.54614	4.060%	1.25625	0.63973	4.200%
δA	0.50080	0.32156	2.675%	0.58615	0.31695	2.975%	0.55800	0.33562	2.670%

improvement is observed over 250 consecutive epochs. The convergence threshold $\varepsilon = 1\times10\text{-}4$ was set to ensure convergence while avoiding excessive computation (Gasteiger et al., 2019). The number of consecutive observation intervals was set to 25 time steps, and the prediction granularity was single-step.

The training processes output the number of training epochs and the loss value. The true values of the three indicators were confirmed in advance. After training, the model's performance was calculated using three metrics: mean squared error (MSE), mean absolute error (MAE), and mean absolute percentage error (MAPE), defined as Equations (18)—(20):

$$\text{MSE} = \frac{\sum_i^n (y_i - \hat{y}_i)^2}{n} \tag{18}$$

$$\text{MAE} = \frac{\sum_i^n |y_i - \hat{y}_i|}{n} \tag{19}$$

$$\text{MAPE} = \frac{\sum_i^n \left|\frac{y_i - \hat{y}_i}{y_i}\right|}{n} \times 100\% \tag{20}$$

where n is the number of samples, y_i is the actual observed value of the ith sample, and $\hat{y}_i$ is the model's predicted value for the ith sample.

The performance metrics are shown in Table 2. The prediction error of A_{sum}, PCI', and δA on the test set was 2.910%, 4.200%, and 2.670%, respectively. The loss curves for training are shown in Figure 6. All losses for three metrics were normalized to have a maximum value of 1, enabling direct comparison across indicators with different original scales. The results show that the model converged quickly during the training process.

Representative cases of cracks and potholes were selected to illustrate the differences between predicted and actual values during testing as shown in Figure 7. The results of indicators PCI' and δA for a pothole case are shown in Figure 7a,b, while the results of indicators PCI' and δA for a cracking case are shown in Figure 7c,d.

Figure 7a presents that the proposed PaveGNet framework that demonstrates a strong fit to the indicator δA under typical conditions. However, when the indicator δA exhibited extreme values, the model's accuracy declined. As shown in Figure 7b, for indicator PCI', the PaveGNet framework underestimated the actual values, reflecting a conservative bias. This may be attributed to missing data points in the training set, which were recorded as mask (zero) to indicate the absence of observation (rather than a true zero value). The real-world pothole case corresponding to Figure 7a is shown in Figure 5a. The pothole area underwent maintenance on October 11, 2022, evolving from one repaired section accompanied by an unrepaired pothole to two distinct patched potholes. This change is correspondingly reflected in the variations observed in Figure 7a,b.

Figure 7c,d display that the overall rate of change for the cracking case was greater than that of potholes, suggesting that cracks tend to evolve more rapidly. Given the higher weight assigned to cracks in the calculation of the indicator PCI', the predicted values (Figure 7d) reflect larger errors. The real-world case corresponding to Figure 7d is shown in Figure 5b. It illustrates that the adjacent right lane of the target section was resurfaced on November 16, 2022. However, due to material discontinuity between the lanes, a new longitudinal crack developed on January 1, 2023. This crack was subsequently sealed on January 2, 2023. Consequently, the local distress evolution exhibited substantial short-term fluctuations as captured in Figure 7c and reflected in Figure 7d. This primarily occurs in cases where distress evolution exhibits abrupt changes, such as sudden increases or decreases in pothole areas or crack propagation caused by extreme weather events or maintenance activities.

Additionally, Figure 8 presents a localized road map displaying the spatial distribution of observed and predicted values on a given date. The localized PCI (PCI') was categorized based on its value as follows: poor (≤55, red), fair (55–70, orange), satisfactory (70–85, yellow), and good (>85, green). Similarly, the distress evolution rate (δA) was classified into severe increase (>5, red), moderate increase

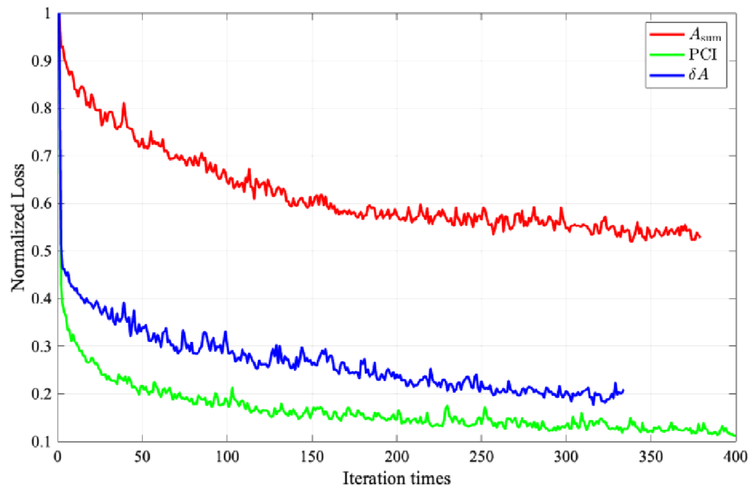

FIGURE 6 Loss curve for training the PaveGNet framework. PCI, pavement condition index.

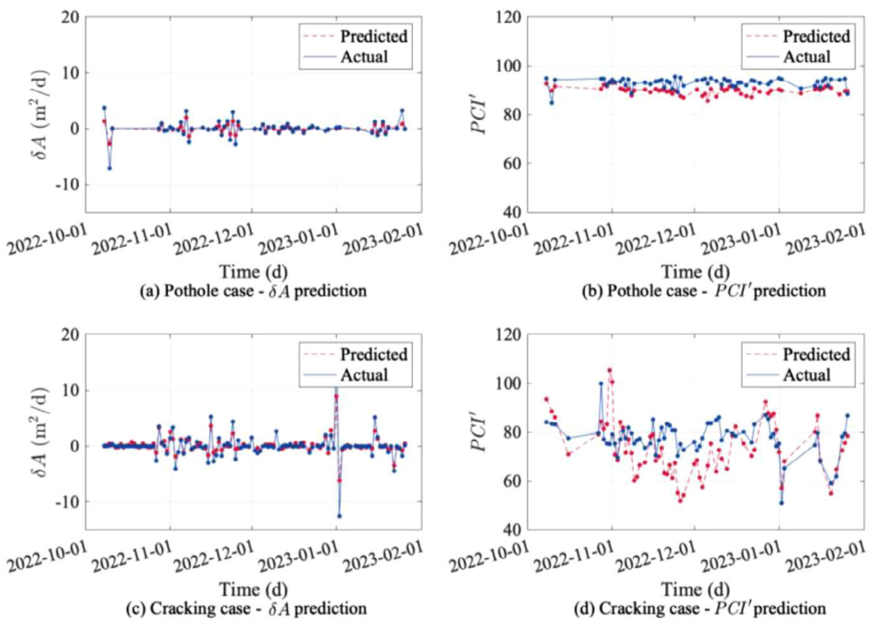

FIGURE 7 Differences between predicted and actual values during testing of the PaveGNet framework.

(1–5, orange), mild increase (0–1, yellow), and repair (<0, green).

As shown in Figures 7d and 8a,b, the cracking case on January 2, 2023, exhibited an observed value within the poor range, while the corresponding predicted value fell within the fair range. Although the PaveGNet prediction showed a slight deviation, the difference between the predicted and actual values was relatively small (57.5659 vs. 50.9747). Therefore, in practical applications, such predictions can still serve as an effective early warning indicator.

The physical location of the pothole and the cracking case is marked. It can be observed that, compared with the pothole case, the crack case was surrounded by more distress instances. Such spatial clustering of distresses likely increases the complexity of the constructed graph, which in turn contributes to larger prediction errors.

Overall, the predicted values of both PCI' and δA show good agreement with the observed data. Larger prediction errors occasionally appear in regions with highly clustered distresses, suggesting that the PaveGNet framework effectively captures the evolution of pavement distresses and enables the differentiation of risk levels, thereby supporting more informed repair planning.

5 | DISCUSSIONS

The framework is designed with a modular architecture, enabling component-wise replacement for broader applicability. To evaluate the modular robustness of the proposed framework, we conducted a series of controlled experiments.

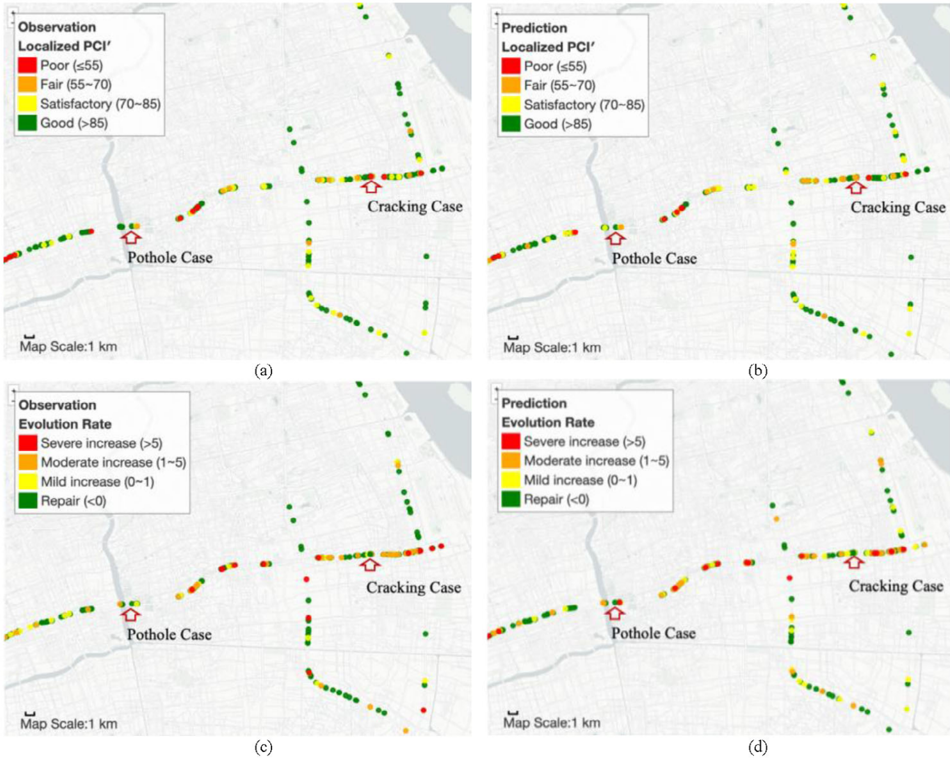

FIGURE 8 Differences between predicted and actual values of the PaveGNet framework on a local road network map (Date: 2023-01-02).

5.1 | Comparative experiments with baseline models

To evaluate the performance of the proposed PaveGNet relative to baseline models, we conducted ablation and comparative experiments. All models were trained and evaluated using the same input features and dataset splits for training, validation, and testing (7:1.5:1.5 ratio). Specifically, we compared PaveGNet with baseline models including Simple RNN, LSTM, Transformer, and STGCN models. All models were trained under consistent hyperparameters: input feature dimension of 12, hidden dimension of 64, and the output dimension of 1; the number of Transformer layers was set to 2, with eight attention heads. Models were trained using the Adam optimizer with a learning rate of 0.001, a batch size of 64, a dropout of 0.2, a maximum of 1000 epochs, and an early stopping patience of 250. Performance metrics, including MSE, MAE, and MAPE, were computed on the three target indicators and are summarized in Table 3.

As shown in Table 3, the proposed PaveGNet consistently outperforms all baseline models, including RNN, LSTM, Transformer, and STGCN, across all three predicted indicators.

For δA, which characterizes the temporal variation in distress, PaveGNet achieved the lowest testing errors (MSE = 0.55800, MAE = 0.33562, MAPE = 2.670%). This represents a substantial improvement compared with the best-performing baseline, STGCN (MSE = 2.25966, MAE = 1.14596, MAPE = 9.005%), indicating that the proposed framework effectively enhances sensitivity to localized distress changes while maintaining numerical stability.

For PCI', PaveGNet achieved MSE = 1.25625, MAE = 0.63973, and MAPE = 4.200%, outperforming all baseline models. Compared with the STGCN model (MSE = 1.70267, MAE = 1.16765, MAPE = 4.703%), the improvement is relatively modest. This suggests that for large-scale or more stable prediction targets such as PCI', the enhancement over existing state-of-the-art STGCN architectures is less pronounced than for finer-grained indicators like A_{sum} and δA, where temporal–spatial interactions and physical constraints play a more critical role.

For A_{sum}, which captures overall distress accumulation, PaveGNet again demonstrates superior predictive accuracy (MSE = 0.13119, MAE = 0.15157, MAPE = 2.910%), substantially lower than those of Transformer (MSE = 0.36374, MAE = 0.18103, MAPE = 3.886%) and STGCN (MSE = 0.32534, MAE = 0.17033, MAPE = 3.701%). This indicates that PaveGNet's spatial–temporal learning structure enables it to model cumulative degradation

TABLE 3 Results of the comparative experiments with baseline models.

	Recurrent neural network			Long short-term memory network			Transformer			Spatial–temporal graph convolutional network		
Predicted indicators	Testing MSE	Testing MAE	Testing MAPE/%	Testing MSE	Testing MAE	Testing MAPE/%	Testing MSE	Testing MAE	Testing MAPE/%	Testing MSE	Testing MAE	Testing MAPE/%
A_{sum}	0.52213	0.22579	5.143%	0.43569	0.19117	4.107%	0.36374	0.18103	3.886%	0.32534	0.17033	3.701%
PCI'	2.35956	1.37637	5.536%	2.22312	1.35436	5.513%	2.75677	1.48265	5.923%	1.70267	1.16765	4.703%
δA	3.30583	1.55818	12.087%	3.59448	1.44566	10.987%	3.02102	1.39119	10.942%	2.25966	1.14596	9.005%

TABLE 4 Results of the ablation study of PaveGNet components.

	Without Temporal interdependence			Without spatial correlation			Without external variables		
Predicted indicators	Testing MSE	Testing MAE	Testing MAPE/%	Testing MSE	Testing MAE	Testing MAPE/%	Testing MSE	Testing MAE	Testing MAPE/%
A_{sum}	0.13812	0.13781	2.740%	0.13619	0.13187	2.777%	0.20193	0.15637	2.946%
PCI'	4.01150	1.89246	7.960%	3.42900	1.71819	7.180%	3.87350	1.80909	8.120%
δA	1.04422	0.55049	3.850%	1.68270	0.79832	6.080%	1.46390	0.75279	6.110%

processes more effectively than sequential or simple spatiotemporal networks.

Overall, the experimental results demonstrate that PaveGNet surpasses all baseline architectures by a considerable margin in both absolute and relative error metrics. The performance gains are attributed to its adaptive spatial–temporal graph learning and integration of exogenous variables, which together enhance the model's ability to capture complex nonlinear deterioration dynamics beyond what conventional RNN, LSTM, Transformer, or STGCN-based methods can achieve.

5.2 | Ablation study of PaveGNet components

To quantify the contribution of each component in the proposed PaveGNet framework, we conducted ablation experiments under three controlled conditions: (i) spatial information only, where temporal dependencies and external variables were excluded; (ii) temporal information only, where spatial correlations and external variables were removed; and (iii) without external variables, where only spatial–temporal features were retained. All experiments used the same data splits and training settings as the full model. The results are summarized in Table 4.

Compared with the full model (see Table 2), removing temporal interdependence increased the MSE for predicting δA to 1.04422, MAE to 0.55049, and MAPE to 3.850%, highlighting the critical role of temporal dependencies in capturing distress evolution. Excluding spatial correlation yielded a similar degradation (MSE = 1.68270, MAE = 0.79832, MAPE = 6.080%), indicating that spatial relationships among distresses are essential for accurate prediction. Removing external variables caused the largest performance drop (MSE = 1.46390, MAE = 0.75279, MAPE = 6.110%), demonstrating that exogenous factors such as weather and traffic conditions are important.

It is noteworthy that although the without external variables case shows a slightly higher MAPE (6.110%) than the without spatial correlation case (6.080%), its MSE and MAE are smaller. This discrepancy arises because MAPE is more sensitive to small-valued samples when the true target values are close to zero. Therefore, the higher MAPE observed in the without external variables case indicates that the model performs poorly in predicting small-magnitude distress changes, suggesting that the inclusion of external factors helps capture subtle variations in distress deterioration more effectively.

For PCI', the trend is consistent: removing each component led to increased errors, with external variables contributing most significantly to overall predictive accuracy (full model: MSE = 1.25625, MAE = 0.63973, MAPE = 4.200%; without external variables: MSE = 3.87350, MAE = 1.80909, MAPE = 8.120%).

For A_{sum}, it can be observed that, since the outputs are linearly related to the inputs, removing each component had a relatively minor impact on the final prediction. In all three cases, the MAPE results remain close to 2.910% as in the full model; however, when external variables are excluded, the MSE increases noticeably to 0.20193.

Overall, the results indicate that the removal of any single component leads to a degradation in performance, with the absence of external variables producing the most pronounced negative impact across all evaluation metrics.

5.3 | Component replacement experiments

In addition to the ablation experiments that remove specific components (e.g., temporal or spatial modules), we also conducted component replacement experiments to further assess the contribution of our architectural design.

The component replacement strategy serves two key purposes. First, it assesses the modular independence and robustness of the proposed framework by verifying whether alternative architectures can maintain or improve performance. Second, it reduces the risk of overfitting to specific learning modules, ensuring that the framework does not rely on any single architecture to perform effectively.

Specifically, the first modification involved constructing a graph attention network (GAT; Veličković et al., 2017) to replace the GCN structure in the original model. Unlike GCNs, which aggregate information uniformly from neighboring nodes, GATs introduce learnable attention mechanisms that assign different weights to each neighbor, thereby enabling the model to prioritize more influential spatial interactions. The principle of GAT is given by Equations (21) and (22):

$$h_v^{t+1} = \rho\left(\sum_{u\in\mathcal{N}_v} \alpha_{vu} W h_u^t\right) \tag{21}$$

$$\alpha_{vu} = \frac{\exp(LeakyReLU(a^T[Wh_v||Wh_u]))}{\sum_{k\in\mathcal{N}_u}\exp(LeakyReLU(a^T[Wh_v||Wh_k]))} \tag{22}$$

where α_{vu} represents the attention score of neighbouring nodes u to the central node v, W is the weight matrix associated with the linear transformation of each node, and a^T is the weight parameter of the attention output. To further stabilize the attention calculation process, multi-head attention was introduced in GAT as shown in Equations (23) and (24):

$$h_v^{t+1} = ||_{k=1}^{K}\sigma\left(\sum_{u\in\mathcal{N}_v} \alpha_{vu}^k W_k h_u^t\right) \tag{23}$$

$$h_v^{t+1} = \sigma\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{u\in\mathcal{N}_v} \alpha_{vu}^k W_k h_u^t\right) \tag{24}$$

where α_{vu}^k is the normalized attention score calculated by the kth attention head. In this study, eight attention heads were adopted.

The performance of the proposed PaveGNet framework after replacing GCN with GAT is shown in Table 5. The results show that the GAT module had a good effect on improving the predictions of A_{sum} and PCI', with MAPE values on the testing set reaching 2.747% and 4.120%, respectively. However, for the prediction of δA, the model showed an error of 5.150%, which was an increase of 2.480%. Specifically, the GAT introduces additional learnable attention coefficients for each pair of connected nodes, substantially increasing the model's parameter count and flexibility. While this enhances its ability to capture complex spatial heterogeneity, it also makes the model more sensitive to local noise when the dataset is relatively small or imbalanced. In our dataset, the δA indicator corresponds to the short-term variation rate of individual distresses, which exhibits higher variability and a lower signal-to-noise ratio than the cumulative indicators (A_{sum} and PCI'). This high temporal volatility amplifies the influence of attention weight fluctuations, causing the GAT-based model to overfit localized temporal patterns rather than generalizable spatial dependencies.

The temporal self-attention network (TSAN) was also introduced to replace the 1-D Conv layer in the framework. The TSAN time learning network is a deep learning model used to process sequential data, especially time-dependent sequential data. TSAN is based on the self-attention mechanism, allowing the model to capture the relationships and dependencies between different time steps in the sequence.

The TSAN leverages self-attention mechanisms to capture temporal dependencies in sequential data. By assigning attention weights to each time step, the model learns the relative importance and correlation of observations across time. The use of multi-head self-attention further enhances this capability by enabling the parallel extraction of diverse temporal patterns. TSAN typically comprises stacked self-attention layers, optionally combined with convolutional or FC layers, to learn rich feature representations. Optimized via backpropagation, the model is trained to minimize a task-specific loss function. Its ability to model long-range temporal dependencies makes TSAN effective for time-series tasks.

Temporal dependencies are modeled using multi-head self-attention with eight attention heads as well to maintain comparability and a model dimension of 64, the same as the pre-defined temporal channels. A feedforward layer with 128 hidden units (the same as the pre-defined hidden channels) and a dropout rate of 0.2 was also applied following each attention block. The attention mechanism operates over a local temporal window of 25 time steps, the same as our PaveGNet.

As shown in Table 6, the results indicate that the TSAN module improved the prediction of A_{sum}. However, it did not provide a notable improvement for the prediction of PCI' and δA. Although there was no overfitting, the computational complexity of the TSAN module was generally higher. The computation time was about twice

TABLE 5 Performance evaluation of the proposed PaveGNet framework (graph attention network replaces graph convolutional network).

Predicted indicators	Training			Validation			Testing		
	Loss	MAE	MAPE/%	MSE	MAE	MAPE/%	MSE	MAE	MAPE/%
A_{sum}	0.07573	0.10794	2.279%	0.08788	0.13078	2.321%	0.18171	0.13340	2.747%
PCI'	1.07795	0.61927	3.850%	1.18080	0.61336	3.780%	1.34115	0.67453	4.120%
δA	0.82778	0.58238	4.850%	0.83050	0.53615	4.940%	1.09340	0.59582	5.150%

TABLE 6 Performance evaluation of the proposed PaveGNet framework (temporal self-attention network replaces 1-D Conv).

Predicted indicators	Training			Validation			Testing		
	Loss	MAE	MAPE/%	MSE	MAE	MAPE/%	MSE	MAE	MAPE/%
A_{sum}	0.16099	0.14493	2.401%	0.10973	0.15214	2.547%	0.19719	0.14098	2.563%
PCI'	1.46321	0.64631	3.880%	1.23735	0.65134	4.230%	1.63054	0.71487	4.710%
δA	0.80930	0.51984	4.470%	0.85125	0.52019	4.820%	0.92063	0.55060	4.170%

that of the previous models. In addition, the improvement for predicting A_{sum} is not significant. Overall, the original PaveGNet model performs better.

These ablation results provide empirical support for the modular design philosophy underpinning the proposed PaveGNet framework. Replacing the original GCN with GAT demonstrated performance gains in predicting key distress indicators such as A_{sum} and PCI', where the MAPE was notably reduced. This indicates that the framework can accommodate alternative spatial learning modules without compromising, and in some cases enhancing, predictive accuracy. However, the observed poor performance in δA prediction highlights that such improvements are not uniformly distributed across all indicators. Similarly, substituting the temporal module with TSAN yielded improved predictions for A_{sum}. However, its limited performance on PCI' and δA and substantially increased computational cost to converge reveal a trade-off between temporal modeling precision and efficiency. These findings further validate the framework's adaptability, demonstrating that individual modules can be tailored for specific prediction targets or operational constraints.

In summary, the replacement experiments confirm the robustness, flexibility, and extensibility of the PaveGNet framework, making it suitable for forecasting diverse pavement distress indicators while mitigating the risk of overfitting to any single component. Together, these evaluations reinforce the credibility of the modular design and its potential for generalizable deployment.

6 | CONCLUSION

This study presents a PaveGNet framework, a spatiotemporal graph network architecture to address the limitations of traditional prediction models on spatiotemporal prediction of pavement distress evolution in road networks, particularly their inability to capture complex spatiotemporal dependencies and network-wide interactions.

By extending the state space from individual distresses to a graph-based multi-node representation, the proposed framework effectively modeled distress evolution as a coupled spatiotemporal process, with exogenous factors such as temperature, precipitation, traffic volume, and maintenance activities embedded as dynamic inputs. The proposed framework includes temporal gated convolution layers and spatial gated convolution layers, which capture the cumulative effect of distress development in the temporal dimension and the spatial correlation. Experimental evaluations demonstrated the model's strong predictive performance, with significant improvements in prediction accuracy. Notably, the prediction error of distress evolution rate δA on the testing set decreased from 9.005% (baseline STGCN model) to 2.670%.

Furthermore, the ablation study demonstrates that the temporal interdependence, spatial correlation, and external variables components of the proposed framework are all important, with the external variables being the most critical. Removing the external variables increases the prediction errors to 2.946%, 8.120%, and 6.110% for indicators A_{sum}, PCI', and δA, respectively. The component replacement experiments also show that the proposed framework is highly modular, allowing each component to be replaced or adapted to different application scenarios.

Overall, the proposed PaveGNet framework offers a more accurate, scalable, and data-driven solution for pavement distress evolution prediction. By explicitly modeling spatial dependencies and temporal dynamics, this approach filled in the gap in predicting short-term spatiotemporal prediction of pavement distress evolution. Future work will focus on further improving model generalizability by extending validation across different geographic regions, incorporating transfer learning, and considering additional physical and functional relationships among road segments. Future work will also explore extending PaveGNet with probabilistic modeling (Xie et al., 2026) and hybrid mechanistic-data-driven approaches (Y. X. Wu et al., 2025) to quantify prediction uncertainty and enhance physical interpretability, thereby providing more actionable insights for maintenance decision-making. In addition, inspired by recent advances in machine learning, alternative and complementary learning paradigms such as the finite element machine for fast learning (Pereira et al., 2020) and self-supervised learning frameworks (Rafiei et al., 2022) will be explored to enhance model adaptability and reduce the reliance on labeled data. These directions will contribute to a more robust, efficient, and generalizable modeling framework for pavement management and infrastructure health monitoring (Javadinasab Hormozabad et al., 2021).

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grant Number: 52372305), Scientific Research Innovation Program of Shanghai Municipal Education Commission (Grant Number: 2021-01-07-00-07-E00092), and UK Research and Innovation (UKRI) and Engineering and Physical Sciences Research Council (EPSRC) (Research Hub for Decarbonised Adaptable and Resilient Transport Infrastructures (DARe), Grant Number: EP/Y024257/1). The authors are responsible for all views and opinions expressed in this paper.

REFERENCES

Abaza, K. A. (2016). Back-calculation of transition probabilities for Markovian-based pavement performance prediction models. *International Journal of Pavement Engineering*, *17*(3), 253–264. https://doi.org/10.1080/10298436.2014.993185

Abaza, K. A. (2017). Empirical approach for estimating the pavement transition probabilities used in non-homogenous Markov chains. *International Journal of Pavement Engineering*, *18*(2), 128–137. https://doi.org/10.1080/10298436.2015.1039006

Abaza, K. A., Ashur, S. A., Abu-Eisheh, S. A., & Rabay'a, A. (2001). Macroscopic optimum system for management of pavement rehabilitation. *Journal of Transportation Engineering-ASCE*, *127*(6), 493–500. https://doi.org/10.1061/(ASCE)0733-947X(2001)127:6(493)

Ahmadlou, M., & Adeli, H. (2017). Complexity of weighted graph: A new technique to investigate structural complexity of brain activities with applications to aging and autism. *Neuroscience Letters*, *650*, 103–108. https://doi.org/10.1016/j.neulet.2017.04.009

Alam, K. M. R., Siddique, N., & Adeli, H. (2020). A dynamic ensemble learning algorithm for neural networks. *Neural Computing & Applications*, *32*(12), 8675–8690. https://doi.org/10.1007/s00521-019-04359-7

Al Aryani, M., Boussabaine, H., & Kirkham, R. (2018). Markov chain optimisation for pavement maintenance. In S. Fırat, J. Kinuthia, & A. Abu-Tair (Eds.), *Lecture notes in civil engineering: Vol. 1. Proceedings of 3RD International sustainable buildings symposium (ISBS 2017)* (pp. 689–704). Springer.

Amezquita-Sanchez, J. P., Valtierra-Rodriguez, M., & Adeli, H. (2018). Wireless smart sensors for monitoring the health condition of civil infrastructure. *Scientia Iranica*, *25*(6), 2913–2925. https://doi.org/10.24200/sci.2018.21136

Askarzadeh, T., Bridgelall, R., & Tolliver, D. D. (2025). Drones for road condition monitoring: Applications and benefits. *Journal of Transportation Engineering Part B-Pavements*, *151*(1). https://doi.org/10.1061/JPEODX.PVENG-1559

Bianchini, A., & Bandini, P. (2010). Prediction of pavement performance through neuro-fuzzy reasoning. *Computer-Aided Civil and Infrastructure Engineering*, *25*(1), 39–54. https://doi.org/10.1111/j.1467-8667.2009.00615.x

Bruna, J., Zaremba, W., Szlam, A., & LeCun, Y. (2013). *Spectral networks and locally connected networks on graphs*. arXiv preprint arXiv:1312.6203. https://doi.org/10.48550/arXiv.1312.6203

Chen, Z., & Ho, P.-H. (2019). Global-connected network with generalized ReLU activation. *Pattern Recognition*, *96*, 106961. https://doi.org/10.1016/j.patcog.2019.07.006

Chu, C.-Y., & Durango-Cohen, P. L. (2007). Estimation of infrastructure performance models using state-space specifications of time series models. *Transportation Research Part C*, *15*, 17–32. https://doi.org/10.1016/j.trc.2006.11.004

Dao, D. V., Adeli, H., Ly, H. B., Le, L. M., Le, V. M., Le, T. T., & Pham, B. T. (2020). A sensitivity and robustness analysis of GPR and ANN for high-performance concrete compressive strength prediction using a Monte Carlo simulation. *Sustainability*, *12*(3), 830. https://doi.org/10.3390/su12030830

Deng, Y., & Shi, X. M. (2024). Short-term predictions of asphalt pavement rutting using deep-learning models. *Journal of Transportation Engineering Part B-Pavements*, *150*(2). https://doi.org/10.1061/JPEODX.PVENG-1409

Du, X., Cai, Y., Wang, S., & Zhang, L. (2016). Overview of deep learning. *2016 31st Youth Academic Annual Conference of Chinese Association of Automation (YAC)*, Wuhan, China (pp. 159–164). https://doi.org/10.1109/YAC.2016.7804882

Du, Y. C., Pan, N., Xu, Z. H., Deng, F. W., Shen, Y., & Kang, H. (2021). Pavement distress detection and classification based on yolo network. *International Journal of Pavement Engineering*, *22*(13), 1659–1672. https://doi.org/10.1080/10298436.2020.1714047

Finlay, C., Jacobsen, J.-H., Nurbekyan, L., & Oberman, A. (2020). How to train your neural ode: The world of Jacobian and kinetic regularization. *International Conference on Machine Learning, PMLR*, Virtual (pp. 3154–3164).

Foundation, R. (2025). *Mobility—Motoring FAQs*. https://www.racfoundation.org/motoring-faqs/mobility

Gasteiger, J., Weißenberger, S., & Günnemann, S. (2019). Diffusion improves graph learning. *Advances in Neural Information Processing Systems*, *32*, Vancouver, BC, Canada.

Gomes, P., Rossi, S., & Toni, L. (2021). Spatio-temporal Graph-RNN for point cloud prediction. *2021 IEEE International Conference on Image Processing (ICIP)*, Anchorage, AK (pp. 3428–3432). https://doi.org/10.1109/ICIP42928.2021.9506084

Gouk, H., Frank, E., Pfahringer, B., & Cree, M. J. (2021). Regularisation of neural networks by enforcing Lipschitz continuity. *Machine Learning*, *110*(2), 393–416.

Haddad, A. J., Saad, G. A., & Chehab, G. R. (2023). Kalman filter updating of rutting predictive models in flexible pavements using measured field data. *International Journal of Pavement Engineering*, *24*(2), 2058700. https://doi.org/10.1080/10298436.2022.2058700

Han, C., Ma, T., Gu, L., Cao, J., Shi, X., Huang, W., & Tong, Z. (2022). Asphalt pavement health prediction based on improved transformer network. *IEEE Transactions on Intelligent Transportation Systems*, *24*(4), 4482–4493. https://doi.org/10.1109/TITS.2022.3229326

Harvey, J., Al-Qadi, I. L., Ozer, H., & Flintsch, G. (2020). *Pavement, roadway, and bridge life cycle assessment 2020: Proceedings of the international symposium on pavement. roadway, and bridge life cycle assessment 2020 (LCA 2020, Sacramento, CA, 3-6 June 2020)*. CRC Press.

Hewamalage, H., Bergmeir, C., & Bandara, K. (2021). Recurrent neural networks for time series forecasting: Current status and future directions. *International Journal of Forecasting*, *37*(1), 388–427. https://doi.org/10.1016/j.ijforecast.2020.06.008

Hu, Y. Y., Chen, N., Hou, Y., Lin, X. S., Jing, B. H., & Liu, P. F. (2025). Lightweight deep learning for real-time road distress detection on mobile devices. *Nature Communications*, *16*(1), 4212. https://doi.org/10.1038/s41467-025-59516-5

Ilicali, M., Catbas, N., Kiziltas, M., & Ongel, A. (2014). Multimodal transportation issues in Istanbul: A case study for traffic redistribution due to long span bridge rehabilitation. *Advances in Civil Engineering and Building Materials III*, *831*, 413–417.

Javadinasab Hormozabad, S., Gutierrez Soto, M., & Adeli, H. (2021). Integrating structural control, health monitoring, and energy harvesting for smart cities. *Expert Systems*, *38*(8), e12845. https://doi.org/10.1111/exsy.12845

Kaur, D., & Tekkedil, D. (2000). Fuzzy expert system for asphalt pavement performance prediction. *2000 IEEE Intelligent Transportation Systems Proceedings*, Dearborn, MI (pp. 428–433).

Kitaha, A., & Biligiri, K. P. (2017). Prioritization-optimization process algorithm to manage pavement networks during the non-availability of historical data. *Journal of Testing and Evaluation*, *45*(2), 546–563. https://doi.org/10.1520/JTE20150287

Kutz, J. N., Brunton, S. L., Brunton, B. W., & Proctor, J. L. (2016). *Dynamic mode decomposition: Data-driven modeling of complex systems*. SIAM. https://doi.org/10.1137/1.9781611974508

Li, T., & Wu, T. (2023). Modeling nonlinear flutter behavior of long-span bridges using knowledge-enhanced long short-term memory network. *Computer-Aided Civil and Infrastructure Engineering*, *38*(11), 1504–1519. https://doi.org/10.1111/mice.13000

Li, Y., Tarlow, D., Brockschmidt, M., & Zemel, R. (2015). *Gated graph sequence neural networks*. arXiv preprint arXiv:1511.05493. https://doi.org/10.48550/arXiv.1511.05493

Li, Z., Wang, Y., Liang, G., Chen, Y., & Yang, X.-l. (2024). *Research status of long-term skid resistance performance of asphalt pavements*. https://doi.org/10.2139/ssrn.4760287

Liu, F. Y., & Al-Qadi, I. L. (2025). Graph neural network-based 3D asphalt concrete pavement response modelling. *International Journal of Pavement Engineering*, *26*(1), 2464201. https://doi.org/10.1080/10298436.2025.2464201

Liu, Y. C., Liu, F., Huang, Y. C., Hu, J., Zhang, W., & Hou, Y. (2025). The real-time pavement distress detection system based on edge-cloud collaborative computing. *IEEE Transactions on Intelligent Transportation Systems*, *26*, 10512–10522. https://doi.org/10.1109/TITS.2025.3544240

Pan, N., Liu, H., Wu, D. F., Liu, C. L., & Du, Y. C. (2023). Spatiotemporal matching method for tracking pavement distress using high-frequency detection data. *Computer-Aided Civil and Infrastructure Engineering*, *38*(16), 2257–2278. https://doi.org/10.1111/mice.12947

Pereira, D. R., Piteri, M. A., Souza, A. N., Papa, J. P., & Adeli, H. (2020). FEMa: A finite element machine for fast learning. *Neural Computing and Applications*, *32*(10), 6393–6404. https://doi.org/10.1007/s00521-019-04146-4

Perez-Ramirez, C. A., Amezquita-Sanchez, J. P., Valtierra-Rodriguez, M., Adeli, H., Dominguez-Gonzalez, A., & Romero-Troncoso, R. J. (2019). Recurrent neural network model with Bayesian training and mutual information for response prediction of large buildings. *Engineering Structures*, *178*, 603–615. https://doi.org/10.1016/j.engstruct.2018.10.065

Rafiei, M. H., & Adeli, H. (2017). A new neural dynamic classification algorithm. *IEEE Transactions on Neural Networks and Learning Systems*, *28*(12), 3074–3083. https://doi.org/10.1109/TNNLS.2017.2682102

Rafiei, M. H., & Adeli, H. (2018). A novel unsupervised deep learning model for global and local health condition assessment of structures. *Engineering Structures*, *156*, 598–607. https://doi.org/10.1016/j.engstruct.2017.10.070

Rafiei, M. H., Gauthier, L. V., Adeli, H., & Takabi, D. (2022). Self-supervised learning for electroencephalography. *IEEE Transactions on Neural Networks and Learning Systems*, *35*(2), 1457–1471. https://doi.org/10.1109/TNNLS.2022.3190448

Sanchit, A., Arun, G., Vaishnavi, S., & Abhinav, S. (2021). Pavement deterioration analysis for rural roads using HDM-4. *IOP Conference Series: Earth and Environmental Science*, *796*(1), 012023.

Shahini Shamsabadi, S., Aleti, T. R., Birken, R., & Wang, M. (2015). PAVEMAN: A data-driven customizable pavement management system.

Shen, D. H., & Du, J. C., & IEEE. (2004). Grey model for asphalt pavement performance prediction. *ITSC 2004: 7TH International IEEE Conference on Intelligent Transportation Systems*, Washington, DC (pp. 668–672).

Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *The Journal of Machine Learning Research*, *15*(1), 1929–1958.

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., & Bengio, Y. (2017). *Graph attention networks*. arXiv preprint arXiv:1710.10903. https://doi.org/10.48550/arXiv.1710.10903

Wang, K. C. P., Li, Q., Hall, K. D., & Elliott, R. P. (2007). Experimentation with gray theory for pavement smoothness prediction. *Transportation Research Record*, *1990*(1), 3–13. https://doi.org/10.3141/1990-01

Wang, Z., Guo, N., Wang, S., & Xu, Y. (2021). Prediction of highway asphalt pavement performance based on Markov chain and artificial neural network approach. *The Journal of Supercomputing*, *77*, 1354–1376. https://doi.org/10.1007/s11227-020-03329-4

Wu, J. M., & Dai, J. L. (2010). Asphalt pavement performance forecast by using gray system theory. *International Workshop on Energy and Environment in the Development of Sustainable Asphalt Pavements*, Virtual (pp. 311–315).

Wu, X., Ajorlou, A., Wu, Z., & Jadbabaie, A. (2023). Demystifying oversmoothing in attention-based graph neural networks. *Advances in Neural Information Processing Systems*, *36*, New Orleans, LA (pp. 35084–35106).

Wu, Y. X., Yin, Z. P., Gao, Y. F., Yang, S. C., & Hou, Y. (2025). Constitutive model-constrained physics-informed neural networks framework for nonlinear structural seismic response prediction. *Computer Methods in Applied Mechanics and Engineering*, *443*, 118079. https://doi.org/10.1016/j.cma.2025.118079

Xiao, F., Chen, X. Y., Cheng, J. C., Yang, S. X., & Ma, Y. (2023). Establishment of probabilistic prediction models for pavement deterioration based on Bayesian neural network. *International Journal of Pavement Engineering*, *24*(2), 2076854. https://doi.org/10.1080/10298436.2022.2076854

Xie, S., Cheng, W., Nie, Z., Huang, Q., Xing, J., Chen, X., Zhang, R., & Yang, Y. (2026). Bayesian physics-informed neural networks with iterative ensemble Kalman inversion for RUL prediction and uncertainty quantification. *Advanced Engineering Informatics*, *69*, 103907. https://doi.org/10.1016/j.aei.2025.103907

Xin, J. Y., Frangopol, D. M., Akiyama, M., & Han, X. (2023). Probabilistic life-cycle connectivity assessment of transportation networks using deep learning. *Journal of Bridge Engineering*, *28*(9). https://doi.org/10.1061/JBENF2.BEENG-6149

Yang, G. W., Li, Q. J., Zhan, Y. J., Wang, K. C. P., & Wang, C. H. (2018). Wavelet based macrotexture analysis for pavement friction prediction. *KSCE Journal of Civil Engineering*, *22*(1), 117–124. https://doi.org/10.1007/s12205-017-1165-x

Yang, J., Lu, J. J., & Gunaratne, M. (2003). *Application of neural network models for forecasting of pavement crack index and pavement condition rating*. https://trid.trb.org/View/643210

Yang, R. K., Liu, L. P., Sun, L. J., Jin, T., Cheng, H. L., & Yuan, J. (2025). Effective temperature model for rutting prediction considering temperature distribution inside the asphalt pavements. *Road Materials and Pavement Design*. Advance online publication. https://doi.org/10.1080/14680629.2025.2477312

Yang, X., Zhang, J. Q., Liu, W. B., Jing, J. Y., Zheng, H., & Xu, W. (2024). Automation in road distress detection, diagnosis and treatment. *Journal of Road Engineering*, *4*(1), 1–26. https://doi.org/10.1016/j.jreng.2024.01.005

Yang, Z. Y., Mohammadi, M., Wang, H. L., & Tsai, Y. C. (2025). A feature-based pavement image registration method for precise pavement deterioration monitoring. *Computer-Aided Civil and Infrastructure Engineering*, *40*(10), 1276–1294. https://doi.org/10.1111/mice.13407

Zhang, H., Chen, N., Li, M., & Mao, S. (2024). The crack diffusion model: An innovative diffusion-based method for pavement crack detection. *Remote Sensing*, *16*(6), 986. https://doi.org/10.3390/rs16060986

Zhang, J., Kong, X., Han, D., & Wang, C. (2016). Prediction model of asphalt pavement PCI of expressway under different maintenance conditions. *Journal of Beijing University of Technology*, *42*(1), 74–80.

Zhang, J. Q., Yang, X., Wang, W., Guan, J. C., Liu, W. B., Wang, H. N., Ding, L., & Lee, V. C. S. (2024). Cross-entropy-based adaptive fuzzy control for visual tracking of road cracks with unmanned mobile robot. *Computer-Aided Civil and Infrastructure Engineering*, *39*(6), 891–910. https://doi.org/10.1111/mice.13108

Zhou, J., Cui, G. Q., Hu, S. D., Zhang, Z. Y., Yang, C., Liu, Z. Y., Wang, L. F., Li, C. C., & Sun, M. S. (2020). Graph neural networks: A review of methods and applications. *AI Open*, *1*, 57–81. https://doi.org/10.1016/j.aiopen.2021.01.001

How to cite this article: Pan, N., Du, Y., & Parlikad, A. K. (2025). A spatiotemporal prediction method for the evolution of pavement distress in road networks. *Computer-Aided Civil and Infrastructure Engineering*, *40*, 6513–6532. https://doi.org/10.1111/mice.70169